VISHNU INSTITUTE OF TECHNOLOGY (AUTONOMOUS)
VISHNUPUR : BHIMAVARAM
MID - II EXAMINATIONS
Advanced Data Structures & Algorithm Analysis
(23CS4T02)

BRANCH : IT & CSBS
YEAR/SEM: II B.TECH II SEM

DATE : 09-04-2026
TIME : 02.20 PM TO 03.50 PM

Max Marks: 30

1		Write an algorithm for job sequencing with deadlines using greedy method and find an optimal job schedule for n=5, profits (P1, P2, P3, P4, P5)=(60, 100, 20 40, 65) and deadlines (d1, d2, d3, d4, d5) =(2, 1, 3, 2, 1)	CO3	L3	[10M]
		(OR)			
2		Solve 0/1 knapsack problem using dynamic programming for knapsack instance n=3, m=6, (p1,p2,p3)=(1,2,5) and (w1,w2,w3)=(2,3,4).	CO3	L3	[10M]
3		Solve 4-Queens problem using back tracking.	CO4	L3	[10M]
		(OR)			
4		Find optimal tour for the following cost matrix using Least cost branch and bound. $\begin{bmatrix} \infty & 5 & 4 & 7 \\ 6 & \infty & 2 & 1 \\ 9 & 7 & \infty & 9 \\ 5 & 9 & 8 & \infty \end{bmatrix}$	CO4	L3	[10M]
5	a	Write about P, NP, NP-hard and NP-complete problems and their relationship.	CO5	L2	[5M]
	b	Explain clique decision problem.	CO5	L3	[5M]
		(OR)			
6		Explain chromatic number decision problem and prove that CNDP is NP-hard.	CO5	L3	[10M]

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VISHNU INSTITUTE OF TECHNOLOGY (AUTONOMOUS)
VISHNUPUR : BHIMAVARAM
MID - II EXAMINATIONS
Short Questions
Design Thinking & Innovation
(23ME4P03)

DATE : 11-04-2026

BRANCH : IT, CSBS

TIME : 02.00 PM TO 02.20 PM

YEAR/SEM: II B.TECH II SEM

Max. Marks: 10

Answer all of the following. Each question carries 2 Marks.

1	Explain the role of creativity in transforming ideas into innovation.	CO3	L2	[2M]
2	What are the characteristics of an innovative organization?	CO3	L1	[2M]
3	What is problem formation in product design and why is it important?	CO4	L2	[2M]
4	What are product specifications? Why are they important?	CO4	L2	[2M]
5	List any three business challenges that design thinking can address in organizations.	CO5	L1	[2M]

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VISHNU INSTITUTE OF TECHNOLOGY (AUTONOMOUS)
VISHNUPUR : BHIMAVARAM
MID - II EXAMINATIONS
Design Thinking & Innovation
(23ME4P03)

BRANCH : IT, CSBS
YEAR/SEM: II B.TECH II SEM

DATE : 11-04-2026
TIME : 02.20 PM TO 03.50 PM

Max Marks: 30

1	Explain the concept of innovation and discuss how organizations convert creative ideas into successful innovations. Illustrate your answer with suitable examples.	CO3	L2	[10M]
	(OR)			
2	Explain the importance of teamwork in the innovation process. Describe how multidisciplinary teams contribute to innovative solutions.	CO3	L2	[10M]
3	Explain the concept of product design and describe the stages involved in the product design process.	CO4	L2	[10M]
	(OR)			
4	Discuss the importance of problem formulation in product design and explain the steps involved in identifying design problems.	CO4	L3	[10M]
5	Explain the stages involved in the product design process, starting from problem formation to product specification.	CO5	L2	[10M]
	(OR)			
6	How can a startup use Design Thinking to maintain market relevance in a rapidly changing business environment? Explain step by step.	CO5	L3	[10M]

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VISHNU INSTITUTE OF TECHNOLOGY (AUTONOMOUS)
VISHNUPUR : BHIMAVARAM
MID - II EXAMINATIONS
Short Questions
Software Engineering
(23IT4T01)

BRANCH : IT & CSBS

YEAR/SEM: II B.TECH II SEM

DATE : 10-04-2026

TIME : 02.00 PM TO 02.20 PM

Answer all of the following. Each question carries 2 Marks. Max. Marks: 10

1	Explain classification of design activities.	CO3	L1	[2M]
2	Write the differences between function oriented design and object oriented design.	CO3	L1	[2M]
3	With the help of a diagram explain Jelinski and Moranda Model of reliability.	CO4	L2	[2M]
4	Define statistical testing and explain.	CO4	L2	[2M]
5	What are the different types of software maintenance?	CO5	L1	[2M]

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VISHNU INSTITUTE OF TECHNOLOGY (AUTONOMOUS)
VISHNUPUR : BHIMAVARAM
MID - II EXAMINATIONS
Software Engineering
(23IT4T01)

BRANCH : IT & CSBS
YEAR/SEM: II B.TECH II SEM

DATE : 10-04-2026
TIME : 02.20 PM TO 03.50 PM
Max Marks: 30

1		Define cohesion and coupling. List and explain different types of coupling and cohesion.	CO3	L2	[10M]
		(OR)			
2		With the help of an example explain the DFD model of a system.	CO3	L3	[10M]
3		Define Integration testing. Explain different types of Integration testing with the help of diagrams.	CO4	L2	[10M]
		(OR)			
4	a	List and explain different types of Code review techniques.	CO4	L2	[5M]
	b	Explain briefly about Software Documentation	CO4	L1	[5M]
5	a	With the help of a diagram explain CASE Environment.	CO5	L3	[5M]
	b	Explain CASE support in Software life cycle.	CO5	L1	[5M]
		(OR)			
6		Define Software Maintenance. With the help of diagrams explain software maintenance process models.	CO5	L3	[10M]

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VISHNU INSTITUTE OF TECHNOLOGY (AUTONOMOUS)
VISHNUPUR : BHIMAVARAM
MID - II EXAMINATIONS
Short Questions
Advanced Data Structures & Algorithm Analysis
(23CS4T02)

BRANCH : IT & CSBS | DATE : 09-04-2026

YEAR/SEM: II B.TECH II SEM | TIME : 02.00 PM TO 02.20 PM

Answer all of the following. Each question carries 2 Marks. Max. Marks: 10

1	Write control abstraction for greedy method.	L1	CO3	[2M]
2	Define optimal binary search tree.	L1	CO3	[2M]
3	What is m-colorability decision problem?	L1	CO4	[2M]
4	Define state space tree.	L1	CO4	[2M]
5	Define P and NP class problems.	L1	CO5	[2M]

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VISHNU INSTITUTE OF TECHNOLOGY (AUTONOMOUS)
VISHNUPUR : BHIMAVARAM
MID - II EXAMINATIONS
Short Questions
Probability & Statistics
(23BS4T02)

BRANCH : CSE, IT
YEAR/SEM: II B.TECH II SEM

DATE : 07-04-2026
TIME : 02.00 PM TO 02.20 PM

Answer all of the following. Each question carries 2 Marks. Max. Marks: 10

1	A continuous random variable X has a distribution function $F(x)=\begin{cases} 0 & if\ x \le 1 \\ k(x-1)^4 & if\ 1 < x \le 3 \\ 1 & if\ x > 3 \end{cases}$ Then find the p.d.f f(x) of x?	CO3	L2	[2M]
2	A variate X has the following probability distribution X: -3, 6, 9 P(X=x): 1/6, 1/2, 1/3 Find E(x)	CO3	L2	[2M]
3	Find the value of the finite population correction factor for n=10 and N=100?	CO4	L2	[2M]
4	A random sample of size 100 has standard deviation 5. what can you say about the maximum error with 95% confidence?	CO4	L1	[2M]
5	Define type-I and type-II error.	CO5	L1	[2M]

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VISHNU INSTITUTE OF TECHNOLOGY (AUTONOMOUS)
VISHNUPUR : BHIMAVARAM
MID - II EXAMINATIONS
Short Questions
Operating Systems
(23CS4T01)

BRANCH : IT & CSBS | DATE : 08-04-2026

YEAR/SEM: II B.TECH II SEM | TIME : 02.00 PM TO 02.20 PM

Answer all of the following. Each question carries 2 Marks. Max. Marks: 10

1	What is Deadlock? List the four necessary conditions for deadlock.	CO3	L1	[2M]
2	What is the Critical Section Problem in Operating Systems?	CO3	L1	[2M]
3	What is Thrashing?	CO4	L1	[2M]
4	What is an Access Matrix in protection?	CO5	L2	[2M]
5	What is meant by File System Mounting?	CO5	L2	[2M]

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VISHNU INSTITUTE OF TECHNOLOGY (AUTONOMOUS)
VISHNUPUR : BHIMAVARAM
MID - II EXAMINATIONS
Operating Systems
(23CS4T01)

BRANCH : IT & CSBS
YEAR/SEM: II B.TECH II SEM

DATE : 08-04-2026
TIME : 02.20 PM TO 03.50 PM

Max Marks: 30

1	a	Explain different methods for handling deadlocks in an operating system.	CO3	L2	[5M]
	b	Describe the Deadlock Avoidance technique using Banker's Algorithm with an example.	CO3	L3	[5M]
		(OR)			
2		Explain Mutex Locks and how they are used to solve the critical section problem. Compare mutex locks with semaphores.	CO3	L4	[10M]
3		Explain Allocation of Frames in paging and discuss different frame allocation algorithms.	CO4	L4	[10M]
		(OR)			
4	a	Explain Swapping in memory management and discuss its advantages and disadvantages.	CO4	L2	[5M]
	b	Explain Virtual Memory and discuss Demand Paging and Copy-on-Write techniques.	CO4	L2	[5M]
5		Explain different Directory Structures (Single-level, Two-level, Tree-structured, Acyclic graph) used in file systems.	CO5	L4	[10M]
		(OR)			
6		Explain different File Allocation Methods such as Contiguous, Linked, and Indexed allocation with advantages and disadvantages.	CO5	L4	[10M]

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VISHNU INSTITUTE OF TECHNOLOGY (AUTONOMOUS)
VISHNUPUR : BHIMAVARAM
MID - II EXAMINATIONS
Optimization Techniques
(23IT4T02)

BRANCH : IT, CSE (AI&DS), CSE (AI&ML)

DATE : 06-04-2026

YEAR/SEM: II B.TECH II SEM

TIME : 02.20 PM TO 03.50 PM

Max Marks: 30

1	Solve the transportation problem for the optimal solution Factory I, II, III; Demand 300 200 200 P Q R Available I: 5 4 3 250 II: 8 4 3 300 III: 9 7 5 300	CO3	L3	[10M]
	(OR)			
2	Solve the transportation problem for the optimal solution 10 2 3 5 9 35 5 10 15 2 4 40 15 5 14 7 15 20 20 15 13 25 8 30 20 20 40 10 35	CO3	L3	[10M]
	(OR)			
3	Minimize the $f(x_1, x_2) = x_1 - x_2 + 2{x_1}^2 + 2x_1x_2 + {x_2}^2$ with the starting point (0,0) using univariate method.	CO4	L4	[10M]
	(OR)			
4	Find the minimum of the function $f(x) = x^5 - 5x^3 - 20x + 5$ by Fibonacci method in the interval (0, 5) up to 6 iterations.	CO4	L3	[10M]
5	Use dynamic programming approach to solve the following LPP Maximize $f(x, y) = 5x + 4y$ subject to $2x + y \leq 8, x + 2y \leq 10$ and $x \geq 0, y \geq 0$.	CO5	L3	[10M]
	(OR)			
6	Explain briefly computational procedure in dynamic programming.	CO5	L2	[10M]

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